

TUSHAR SOHAL

Backend Engineer

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Professional Summary

Backend Engineer with 1+ years of experience building production-grade backend systems for defence and enterprise applications. Experienced in REST API development, PostgreSQL/PostGIS optimization, SQLAlchemy, GeoAlchemy2, JWT authentication, RBAC, Redis caching and secure backend architecture. Hands-on experience delivering scalable backend services using FastAPI and Node.js while currently strengthening expertise in Java and Spring Boot.

Technical Skills

Languages: Java, Python, JavaScript, SQL
Backend: FastAPI, Node.js, Express.js, REST APIs
Databases: PostgreSQL, PostGIS, MySQL, MongoDB, SQLite
ORM: SQLAlchemy, GeoAlchemy2
Security: JWT, RBAC, IAM
Caching & Tools: Redis, Docker, Git, Postman, Swagger/OpenAPI, CI/CD
Frontend: React.js, Angular, Next.js

Professional Experience

Web Developer Sep 2025 – Present
UPSALA — FastAPI, PostgreSQL, PostGIS, Angular, Next.js

- Developed secure REST APIs for a defence-grade intelligence platform supporting 800+ internal users across hierarchical military organizations.
- Designed authentication and authorization using JWT, refresh tokens, RBAC and IAM integration for secure enterprise access.
- Built GIS-enabled backend services using PostgreSQL, PostGIS, SQLAlchemy and GeoAlchemy2 for dashboards, reports and operational mapping.
- Optimized PostgreSQL queries using joins, indexing and recursive hierarchical queries, improving backend scalability and response time.
- Implemented Redis caching, backend optimizations and production data migration while ensuring application stability.
- Collaborated across the complete software lifecycle from feature development to production deployment following secure development practices.

Professional Projects

The Kapda Company – Multi-Role Custom Apparel Platform
kapdaco.vercel.app

- Designed and developed a multi-role e-commerce platform supporting customers, sellers, designers and administrators.
- Built REST APIs for authentication, product catalog, order management and user administration.
- Integrated Razorpay payment gateway and implemented Role-Based Access Control (RBAC).
- Developed dynamic product customization allowing users to personalize apparel with images and text.

Lord Krishna Convent School Management Platform (Deployed)
lordkrishnaconventschool.com

- Designed, developed and deployed a full-stack school management platform for Lord Krishna Convent School, Gurugram serving 1,000+ users.
- Developed backend APIs for student management, events, enquiries, complaints and content management.
- Implemented secure role-based access and optimized backend APIs for improved responsiveness.

Library Management Desktop Application (Deployed)

- Designed, developed and deployed a Java desktop application for Lord Krishna Convent School to manage inventory, book issuance, returns and fine calculation.
- Implemented Excel import/export for bulk record management and automated workflows, reducing manual administrative effort by approximately 85%.

Education

Master of Computer Applications	2025
Guru Gobind Singh Indraprastha University	GPA: 8.3/10
Bachelor of Computer Applications	2022
Guru Gobind Singh Indraprastha University	GPA: 8.4/10

Certifications

- Backend Development for .NET Full Stack — Coursera — Board Infinity (2025)
- Frontend Development using React — Coursera — Board Infinity (2025)
- Python Data Structures — Coursera — University of Michigan (2021)